

1. System Overview

The ShopHub E-Commerce Platform is a web-based multi-vendor platform designed to facilitate secure and interactive online shopping for customers, while providing comprehensive store management for sellers, and oversight tools for administrators. The system provides powerful tools for managing product catalogs, processing orders, and tracking sales analytics.

The application follows a layered architecture using the MVC (Model-View-Controller) pattern implemented through a Laravel backend with a modern React frontend powered by Inertia.js.

2. System Objectives

The system aims to:

- Provide an engaging and user-friendly digital marketplace for customers
- Enable sellers to easily create, manage, and distribute their product listings
- Ensure product quality through an Admin Product Approval workflow
- Implement secure shopping carts, checkouts, and order tracking mechanics
- Track sales performance and platform analytics
- Facilitate secure transactions and transparent customer-seller interactions
- Ensure secure authentication and strict role-based access control (RBAC)

3. Technology Stack

Category	Technology
Backend Framework	PHP 8.x+ with Laravel 11.0
Database	MySQL
Frontend Framework	React 18.0
UI Styling	Tailwind CSS and Custom Vanilla CSS
Animations	Framer Motion

Category	Technology
State Management/Routing	Inertia.js
Authentication	Laravel Sanctum / Session-based
Build Tools	Vite and Laravel Vite Plugin
Data Visualization	Chart.js & react-chartjs-2
Icons	Lucide React

4. System Architecture

The system follows a three-layer architecture:

Presentation Layer

Handles the user interface and frontend rendering.

Components:

- React Components
- Tailwind CSS UI Components
- Customer Storefront
- Seller Dashboard
- Admin Dashboard
- Authentication Portals (Customer, Seller, Admin)

Application Layer

Handles business logic and request processing.

Components:

- Controllers (Auth, Product, Cart, Checkout, AdminCrud, SellerDashboard)
- Middleware (Authentication, Role Authorization: role:admin, role:seller)
- Product Approval Service
- Order & Checkout Logic

- **Notification Engine**

Data Layer

Handles persistent data storage.

Components:

- **MySQL Database**
 - **Eloquent ORM Models**
 - **Database Migrations & Seeders**
 - **File Storage System (for product images and shop logos)**
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5. User Roles and Permissions

Customer (User)

Customers can:

- **View and access the product catalog**
- **Filter products by category and search by keyword**
- **Add products to the shopping cart and proceed to checkout**
- **Track personal order status and history**
- **Manage their personal profile and password**
- **Receive and manage system notifications**
- **Leave reviews and ratings on purchased products**

Seller

Sellers can:

- **Create and manage their dedicated Shop Profile (logo, description)**
- **Upload and manage product listings (CRUD operations)**
- **Track specific orders made for their products**
- **View shop performance, earnings, and sales analytics**
- **Update order statuses for fulfillment (processing, shipped)**
- **Receive notifications regarding product approvals and orders**

Admin

Administrators can:

- **Manage all user accounts (customers, sellers, and fellow admins)**
 - **View a system-wide dashboard with total platform statistics**
 - **Review, approve, reject, or flag products submitted by sellers**
 - **Manage global product categories**
 - **Monitor all orders platform-wide**
 - **Send messages and announcements to users**
 - **Verify new seller registrations**
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6. Core Functional Modules

6.1 Authentication Module

Features:

- **Separate User, Seller, and Admin registration and login portals**
- **OTP (One-Time Password) verification for secure registration**
- **Profile and password management**
- **Session management and logout functionality**

Security Mechanisms:

- **Session-based authentication**
 - **CSRF protection and password hashing (Bcrypt)**
 - **Role-based route access restrictions**
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6.2 E-Commerce & Product Module

Functions:

- **Global category management**
- **Product listing creation with specifications, stock, and pricing**
- **Multiple image uploads for products**
- **Product approval workflow (Pending, Approved, Rejected)**
- **Dynamic product filtering and search**

Key Models:

- **Product**

- **ProductImage**
 - **Category**
 - **ProductApproval**
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6.3 Cart & Checkout Module

Functions:

- **Add/remove items to/from the cart**
- **Quantity updates and total calculations**
- **Secure checkout process with shipping address inputs**
- **Payment method selection (COD, GCash, Maya)**

Key Models:

- **Cart**
 - **CartItem**
 - **Order**
 - **OrderItem**
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6.4 Order & Fulfillment Module

Functions:

- **Order generation and tracking numbers**
- **Status tracking (Pending, Confirmed, Processing, Shipped, Delivered)**
- **Segmented views (Sellers see only their items; Admins see all)**

Key Models:

- **Order**
 - **OrderItem**
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6.5 Seller Management Module

Functions:

- **Shop profile creation and customization**
- **Seller invite code validation (optional admin control)**

- Earnings and sales tracking

Key Models:

- SellerProfile
 - User
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6.6 Notification & Review Module

Functions:

- System notifications for order updates and approvals
- Product rating and commenting post-purchase

Key Models:

- Notification
 - Review
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7. Routing Structure

Public Routes

Route	Description
/	Home/Storefront
/products	Product Catalog (/ {slug} for details)
/category/{slug}	Browse by category

Authentication Routes

Route	Description
/login & /register	Customer Auth
/seller/login & /seller/register	Seller Auth

Route	Description
/admin/login & /admin/register	Admin Auth
/logout [POST]	Universal Logout

Customer (Authenticated) Routes

Route	Description
/cart	View Cart (/add, /update, /remove)
/checkout	Checkout Process
/orders	Order History (/order for tracking)
/profile	Profile Management
/notifications	Notifications Hub

Seller Routes (Prefix: /admin/seller)

Route	Description
/dashboard	Seller Analytics
/products	Manage Products
/orders	Fulfillment Dashboard
/settings	Shop Settings

Admin Routes (Prefix: /admin)

Route	Description
/dashboard	Admin Analytics
/users & /sellers	Account Management
/products & /categories	Catalog Control
/product-approvals	Approval Workflow Dashboard
/approve, /reject	Approve/Reject Actions
/orders	Global Order Monitoring
/messages/*	Admin Messaging System

8. Application Workflow

Student Workflow (*Customer Workflow*)

1. Customer registers via OTP or logs in
2. Customer browses products or categories on the storefront
3. Customer adds desired items to the shopping cart
4. Customer proceeds to checkout, enters shipping details, and selects a payment method
5. Customer tracks the order status through their profile
6. Once delivered, the customer leaves a product review

Teacher Workflow (*Seller Workflow*)

1. Seller registers and sets up their Shop Profile
2. Seller uploads new products to the platform
3. Products undergo Admin review (Product Approval Workflow)
4. Once approved, products appear on the public storefront
5. Seller receives order notifications when customers buy their products
6. Seller processes the order and updates the shipping status

7. Seller tracks earnings via their dashboard

Admin Workflow (*Admin Workflow*)

- 1. Admin logs in to the main control panel**
 - 2. Admin reviews pending products submitted by sellers (Approve/Reject)**
 - 3. Admin oversees global platform sales and user statistics**
 - 4. Admin manages categories, verifies new sellers, and handles disputes**
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9. File Structure

Directory	Purpose
app/Http/Controllers	Application controllers
app/Models	Eloquent database models
app/Http/Middleware	Custom middleware (Role checking)
resources/js	React frontend components
resources/js/Pages	Page components (Auth, Customer, Seller, Admin)
resources/css	Stylesheets (Tailwind)
routes/	Route definitions
database/migrations	Database schema migrations
database/seeder	Database seeders
public/	Public assets, uploaded images
config/	Application configuration

Directory

Purpose

storage/

File storage (avatars, logos, product images)

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10. Key Features

Advanced Product Approval Workflow

- Prevents spam or inappropriate items from going live
- Admins have complete control over what sellers can publish

Multi-Vendor Dashboarding

- Separate, dynamic Chart.js powered dashboards
- Admins view platform-wide statistics
- Sellers view shop-specific analytics

OTP Secure Registration

- Enhanced security ensuring verified contact points
- Required for both customers and sellers

Inertia.js SPA Experience

- Fast, seamless browsing without page reloads
- Mimics standalone mobile app experience
- Maintains Laravel's robust backend capabilities

Shopping Cart & Checkout

- Real-time cart updates
 - Multiple payment methods (COD, GCash, Maya)
 - Order tracking and history
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11. Future Improvements

Potential future enhancements include:

- Integration of live Payment Gateways (PayMongo, Xendit, Stripe)
- Real-time chat system between Customers and Sellers

- **Mobile application counterpart (React Native or Flutter)**
- **Advanced logistics API integration (Lalamove, J&T) for automated shipping rates and live GPS tracking**
- **AI-Powered product recommendations for customers**
- **Exportable PDF reporting for Seller and Admin analytics**
- **Wishlist and saved items functionality**
- **Discount and voucher system**
- **Multi-language support**
- **Live customer support chat**